**FACULTY OF SCIENCE – UNIVERSITY OF PERADENIYA**
**DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE**

**END SEMESTER EXAMINATION (SEMESTER I) – 2020/2021**

**ST1031 – STATISTICS APPLICATIONS I (1 CREDIT)**

Time allowed: **TWO Hours** Answer **ALL** questions

**INSTRUCTIONS:** You should prepare a text document (MS – WORD or Writer) including all the necessary outputs, answers and relevant interpretation. Save the file with the name **XXXXXX.doc** or **XXXXXX.odt** where **XXXXXX** represents your registration number (Eg: If the registration number is "**S/19/001**" then the file name should be "**S19001.doc**" or "**S19001.odt**")

Also save the R-studio project with the relevant R-scripts with the name "**XXXXXX.Rproj**" in your home directory. **XXXXXX** represents your registration number. Inside this project, create a separate R-script file for each question including all the R commands and save your R-script files with the file names below.

| Question | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| R-script file | Que1.R | Que2.R | Que3.R | Que4.R |

You are advised to save all your work regularly to avoid losses of data due to power failures.

1. The file ***EPL_2021.csv*** contains the details of all the players who played in the English Premier League (EPL) 2020-2021 season. The dataset contains nine variables as follows.

| **Variable** | **Description** |
|---|---|
| Name | Name of the football player. |
| Club | The football club that the player represented during the EPL |
| Nationality | The nationality of the player. |
| Age | The age of the player. |
| Goals | The number of Goals scored by the player. |
| Assists | The number of times the player has assisted another player in scoring the goal. |
| Passes_Completed | The number of passes that the player accurately passed to his teammate. |
| Penalty_Goals | The number of penalty goals scored by the player out of the penalties attempted. |
| Penalty_Attempted | The number of penalty attempts by the player |

i. (10 Points) Import the ***EPL_2021.csv*** file to your R working directory and rename it as ***PremierLeague***.

ii. (10 Points) Thiago Alcantara, a Spanish (ESP) football player was signed by Liverpool FC when he was 29 years old. He started in 20 games out of a total of 24 matches that he played in, scoring 1 goal, attempting 0 penalties, and having no assists. He attempted 1674 passes with a pass completion rate of 89.5%. Add Thiago's details to the dataset.

iii. (15 Points) Obtain the structure and the summary statistics of all the variables in the provided dataset, and interpret the results.

iv. (10 Points) Create a new dataframe named ***PLClubs*** containing the names of all the clubs in the premier league and the number of players that represent each club.

v. (15 Points) Obtain a summary table that contains the number of players from each Nationality playing for each Club in the EPL. From the summary table, extract the data rows that contain the details of Spanish players (Nationality=ESP) who play for Leeds United.

vi. (15 Points) Obtain the sum of goals scored by each club and visualize them using a horizontal barplot. Add a suitable title, x-axis name and y-axis name to the same plot.

vii. (15 Points) The manager of the Liverpool team is comparing players based on the percentage of completed passes (***Passes_Completed***) during the season. Obtain a summary table that would help him to select the best player.

viii. (10 Points) Create a new logical variable named ***forwards***, which is true if the number of goals scored by a player is greater than 3 and number of assists is greater than 1, and false otherwise.

2. a) (5 Points) Create the following matrix in R.

$$\boldsymbol{M} = \begin{bmatrix} 3 & 2 & -1 \\ 2 & -2 & 4 \\ -1 & 0.5 & -1 \end{bmatrix}$$

i. (5 Points) Find the transpose of the matrix $\boldsymbol{M}$ and store it as matrix $\boldsymbol{N}$.

ii. (5 Points) Using matrix multiplication, multiply the two matrices $\boldsymbol{M}$ and $\boldsymbol{N}$.

iii. (10 Points) Is the matrix $\boldsymbol{M}$ invertible? If so, Find the inverse of the matrix $\boldsymbol{M}$.

iv. (15 Points) Hence, or otherwise, solve the following system of equations.

$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= 1 \\ 2x_1 - 2x_2 + 4x_3 &= -2 \rightarrow \ \boldsymbol{MX} = \boldsymbol{A}, \\ -x_1 + 0.5x_2 - x_3 &= 0 \end{aligned}$$

where $\boldsymbol{X^T} = [x_1 \quad x_2 \quad x_3]$ and $\boldsymbol{A^T} = [1 \quad -2 \quad 0]$.

b) The following table shows the marks obtained by five students for four subjects in the final examination of a particular academic year.

| | Heshan | Thinugi | Shanuli | Devin | Piumi |
|---|---|---|---|---|---|
| Mathematics | 80 | 70 | 50 | 82 | 30 |
| Science | 52 | 86 | 81 | 46 | 91 |
| History | 80 | 58 | 72 | 81 | 74 |
| English | 82 | 89 | 52 | 41 | 78 |

i. (10 Points) Create a $5 \times 4$ empty matrix $\boldsymbol{X}$ and store the marks of each subject column wise.

ii. (10 Points) Add column and row names to matrix ***X*** such that, column names represent subjects and row names represent the student names.

iii. (5 Points) Find the mean marks of each subject and store it in a vector called ***X_bar***.

iv. (10 Points) Find the matrix ***D*** such that $\boldsymbol{D} = \boldsymbol{X}\boldsymbol{X}^{\boldsymbol{T}}$

v. (5 Points) Extract the diagonal elements of matrix ***D*** and store it in a vector called ***X_sq***.

vi. (10 Points) Using the results obtained in parts (iii) and (v), find the variance of each subject and interpret your results. (**Hint**: $var(X) = \frac{1}{n-1}(\sum_{i=1}^{n} X_i^2 - n\bar{X}^2)$)

3. a) Create the below given sequence of vectors using the functions ***seq*** and ***rep*** when necessary.

i. (10 Points) $V1 = \{5, 7, 9, \dots, 15\}$

ii. (10 Points) $V2 = \{25, 25.25, 25.50, \dots, 27\}$

iii. (10 Points) $V3 = \{2, 2, \dots, 2, 5, 5, \dots, 5, 7, 7, \dots, 7\}$, where there are 6 occurrences of 2, 7 occurrences of 5 and 8 occurrences of 7.

iv. (10 Points) $V4 = \{6, 7, 8, 6, 7, 8, \dots, 6, 7, 8\}$, where there are 7 occurrences of the set $\{1, 2, 3\}$

v. (10 Points) Obtain a random sample with 15 observations from vector $V2$, and store it as vector $V5$.

b) (5 Points) Create a data frame named ***Product_Details*** and store the data given below.

| ***Product_ID*** | ***Product*** | ***Price (US Dollars)*** | ***Stock*** |
|---|---|---|---|
| P001 | Laptop | 331 | 75 |
| P002 | Monitor | 189 | 82 |
| P003 | Mouse | 14 | 115 |
| P004 | Keyboard | 69 | 125 |

i. (10 Points) Print all the products with stocks greater than 100 US dollars.

ii. (5 Points) Create a new data frame named ***Manufacturer_Details*** with the following details.

| ***PID*** | ***Manufacturer*** | ***Origin*** |
|---|---|---|
| P001 | DELL | USA |
| P005 | Logitech | India |
| P003 | Meetion | China |
| P004 | DELL | USA |
| P006 | Pro-Tech | USA |

iii. (10 Points) Merge the two data frames ***Product_Details*** and ***Manufacturer_Details*** such that merged dataframe contains only the details of the product IDs available in the first data frame (***Product_Details***).

iv. (10 Points) Merge the two data frames ***Product_Details*** and ***Manufacturer_Details*** such that merged dataframe contains all the Product IDs. Name the merged data set as ***New_Product_Details***.

v. (10 Points) Find the number of missing values in ***New_Product_Details***.

4. a) The employees of a company receive a subsistence allowance of Rs. 10000.00 per month apart from their Basic salary (Rs. $x$). All employees are allowed a maximum of 2 leaves for a month and any extra leaves taken are considered as no-pay leaves. The amount reduced for a single no-pay leave is Rs. $x/30$. For each employee, the taxable income (Basic salary+allowence) is then used to identify the Employees Provident Fund (EPF) obligations of the Employer and the employee as follows.

| | Employee contribution | Employer contribution |
|---|---|---|
| EPF | 8% | 12% |

i. (40 Points) Write an R-function to calculate the total salary of an employee with the EPF amount paid from the employee and the employer separately, using the two input arguments of Basic salary $(x)$ and the total number of leaves taken by the employee $(n)$.

ii. (10 Points) Using the R-function created in part (i), find the salary details of an employee whose basic salary is Rs. 30000.00 and who has taken three leaves in the given month.

b) (50 Points) Write an R function, ***Binom_coef***, using recursion that takes two parameters $n$ and $k$ and returns the value of a Binomial Coefficient $C(n,k) = \frac{n!}{k!(n-k)!}$, where $k \leq n$ and $n, k \geq 1$.
(**Hint**: $C(n, k = 1) = n$ and $C(n,k) = C(n-1, k-1) * (n/k)$)

**- End -**